A Way Out of the Impossibility: The First-Best Screening Rule

ECON 470 — Microeconomics of Artificial Intelligence

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1 Where we are

In the previous handout (*Algorithmic Fairness Definitions and their Incompatibility*) we proved that, once base rates differ across groups, **any two** of independence, separation, and sufficiency imply equal base rates. So the three fairness criteria are pairwise mutually exclusive in any realistic application, and no algorithm can satisfy all three at once.

This handout gives **one way out** of the impossibility, due to Rambachan, Kleinberg, Ludwig & Mullainathan (NBER WP 27111, 2020). The idea is to **stop trying to make the prediction itself "fair"** and instead separate the prediction step from the allocation step. The result is a clean, derivable rule: use *all* information — including group membership G — to predict; use *group-specific thresholds* to allocate.

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2 Setup

There are N candidates indexed by i. Each is described by:

- a true outcome $\theta_i \in \{0, 1\}$ (the "positive" / "negative" state);
- a group membership $G_i \in \{0, 1\}$ (with $G = 1$ denoting the protected group);
- a vector of non-group characteristics X_i.

A social planner observes (G_i, X_i) but **not** θ_i at decision time. She chooses an admissions decision $a_i \in \{0, 1\}$ for each candidate, subject to a capacity constraint:

$$\sum_{i=1}^{N} a_i \;\leq\; K. \tag{Capacity}$$

The planner has equity weights $\psi_g \geq 0$ for each group $g \in \{0, 1\}$. Setting $\psi_0 = \psi_1$ corresponds to no intrinsic group preference ; setting $\psi_1 > \psi_0$ means the planner values outcomes in group 1 more.

3 The social welfare function

The planner ranks rules by the **expected** social welfare they deliver:

$$\text{SWF} \;=\; \sum_{i=1}^{N} \psi_{G_i}\, \theta_i\, a_i, \qquad \mathbb{E}[\text{SWF}] \;=\; \mathbb{E}\left[\sum_{i=1}^{N} \psi_{G_i}\, \theta_i\, a_i\right].$$

Reading the formula: a candidate i contributes to social welfare only if (i) we admit her ($a_i = 1$) *and* (ii) she turns out positive ($\theta_i = 1$). The contribution is scaled by the equity weight ψ_{G_i} on her group.

Notice that no AI "score" appears anywhere. The objective is written in terms of the realized outcome θ_i, the admit decision a_i, and the equity weight ψ_{G_i} — nothing else. The score will emerge as soon as we take the expectation.

4 The first-best problem

The planner's problem is to maximize expected social welfare subject to capacity:

$$\max_{\{a_i\} \in \{0,1\}^N} \; \mathbb{E}\left[\sum_{i=1}^{N} \psi_{G_i}\, \theta_i\, a_i\right] \qquad \text{s.t.} \qquad \sum_{i=1}^{N} a_i \;\le\; K, \tag{FB}$$

where each a_i may depend on the observables (G_i, X_i) but not on the unknown θ_i.

We now solve this. The solution is the cleanest possible: a **group-specific threshold rule on a single score**, where the score itself falls out of the derivation.

Proposition (Rambachan, Kleinberg, Ludwig & Mullainathan, 2020). *The first-best admissions rule for (FB) is the group-specific threshold rule*

$$a_i^* \;=\; \mathbf{1}\{\, s_i \;>\; t(G_i;\, K)\,\},$$

where

$$s_i \;:=\; \Pr[\theta_i = 1 \mid G_i, X_i]$$

is the conditional probability that candidate i is positive, given all observable information about her, and the threshold $t(g; K)$ is group-specific and set so that the capacity constraint binds exactly.

5 Proof

The proof has three steps.

5.1 Step 1. The law of iterated expectations

Since each a_i is a function of the observables (G_i, X_i) but not of θ_i, we can apply the law of iterated expectations:

$$\mathbb{E}\left[\sum_{i=1}^{N} \psi_{G_i}\, \theta_i\, a_i\right] \;=\; \sum_{i=1}^{N} \mathbb{E}\Big[\psi_{G_i}\, a_i\, \mathbb{E}[\theta_i \mid G_i, X_i]\Big] \;=\; \mathbb{E}\left[\sum_{i=1}^{N} \psi_{G_i}\, s_i\, a_i\right],$$

where, because θ_i is binary, we have **defined** the AI score

$$\boxed{s_i \;:=\; \mathbb{E}[\theta_i \mid G_i, X_i] \;=\; \Pr[\theta_i = 1 \mid G_i, X_i].}$$

This is the only place the score enters the derivation, and it enters as a *consequence* of the iterated-expectations step, not as a primitive of the problem. The "AI score" is just the conditional probability that candidate i is positive, given everything we observe about her.

5.2 Step 2. The top-K rule

Conditional on the realized observables $(G_i, X_i)_{i=1}^N$, the planner's problem reduces to

$$\max_{\{a_i\} \in \{0,1\}^N} \;\sum_{i=1}^{N} (\psi_{G_i}\, s_i)\, a_i \qquad \text{s.t.} \qquad \sum_{i=1}^{N} a_i \;\le\; K,$$

which is a 0-1 knapsack with **unit item weights** and item values $\psi_{G_i}\, s_i \ge 0$.

The solution is immediate: **admit the K candidates with the largest values of $\psi_{G_i}\, s_i$**, ties broken arbitrarily. (The constraint binds because all item values are non-negative; the objective is non-decreasing in each a_i.) Call this the **top-K rule on the welfare-weighted score $\psi_{G_i}\, s_i$**.

5.3 Step 3. Group-specific threshold

We now show that the top-K rule on $\psi_{G_i}\, s_i$ is *equivalent* to a group-specific threshold rule on s_i alone.

For each group $g \in \{0, 1\}$, let K_g^* denote the number of slots the top-K rule assigns to group g, so $K_0^* + K_1^* = K$. Within group g, the weight $\psi_{G_i} = \psi_g$ is *constant*, so ranking candidates by $\psi_{G_i}\, s_i$ is equivalent to ranking them by s_i alone. The top-K rule therefore admits exactly the K_g^* candidates in group g with the highest values of s_i — which is to say, candidates whose score exceeds the cutoff

$$t(g;\, K) \;:=\; \text{the } K_g^*\text{-th largest value of } s_i \text{ among candidates in group } g.$$

Hence

$$a_i^* \;=\; \mathbf{1}\{\, s_i \;>\; t(G_i;\, K)\,\},$$

with ties at the threshold handled so capacity binds exactly. ■

5.4 Where does ψ_g enter the solution?

The equity weight ψ_g does not enter directly in the threshold comparison $s_i > t(G_i; K)$. Instead, it determines the **split** of slots across groups: $K_0^* + K_1^* = K$. A larger ψ_1 pushes more slots into the protected group — which lowers the threshold $t(1; K)$ and raises $t(0; K)$. But the comparison itself is just "score vs. group-specific threshold."

This is the formal sense in which **equity preferences modify the admissions rule, not the prediction function.**

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6 Three takeaways

Read directly off the formula $s_i = \Pr[\theta_i = 1 \mid G_i, X_i]$:

1. **Do not blind the algorithm to group membership G_i.** Group is *inside* the conditioning set. Dropping it would replace s_i with the strictly less accurate $\Pr[\theta_i = 1 \mid X_i]$.
2. **Do not remove characteristics X_i.** Same logic: X_i is in the conditioning set; removing any of it makes s_i a worse estimate of $\Pr[\theta_i = 1]$.
3. **Do not impose fairness constraints inside the training procedure.** The score s_i is a conditional probability. It is *uniquely determined* by the joint distribution of (θ, G, X). There are no tunable knobs inside it for an equity preference to act on. Equity preferences enter only through ψ_g, which acts on a_i at the allocation stage.

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7 The equity-irrelevance result

From RKLM's abstract:

> *When a social planner builds the algorithm herself, her equity preference has no effect on the training procedure. So long as the data, however biased, contain signal, they will be used and the learning algorithm will be the same. Equity preferences alone provide no reason to alter how information is extracted from data — only how that information enters decision-making.*

In one line:

If you want a fairer outcome, change *how the score is used*, not *how it is made*.

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8 Connection back to the impossibility

The impossibility theorem from the previous handout said: when base rates differ across groups, no algorithm can satisfy any two of {independence, separation, sufficiency} on the *prediction* function. The RKLM first-best rule will, in general, satisfy **none** of the three — and that is fine. The prediction function isn't a *choice* in the first place. It is the conditional probability $\Pr[\theta_i = 1 \mid G_i, X_i]$, determined by the joint distribution of (θ, G, X). The *choice* lives at the allocation stage, in the group-specific threshold $t(g; K)$ — and that is where the planner's equity preference ψ_g has been put to work.

The impossibility theorem was telling us not "no fair algorithm exists" but rather "the prediction function is the wrong place to look for fairness." RKLM's first-best result is the cleanest possible statement of where to look instead.